# Basic Principles of Medicine 2
# Module : Digestion and Metabolism
# Lecture No: 14

## Title: Metabolism of Fructose and Galactose

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# Objectives: Metabolism of Fructose and Galactose

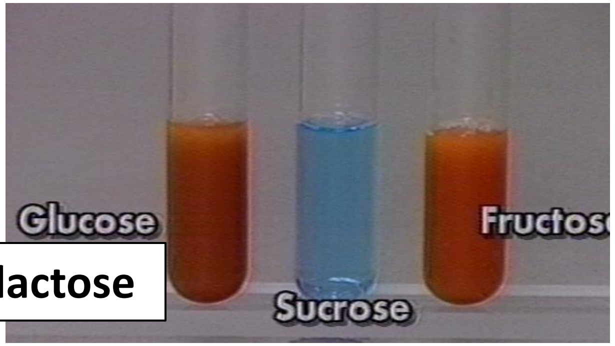
| | |
|---|---|
| SOM. MK.I.BPM2.3.DM.2.BCHM.1167 | Describe the pathway by which dietary fructose enters glycolysis highlighting the role of fructokinase and Aldolase B. |
| SOM. MK.I.BPM2.3.DM.2.BCHM.1168 | Compare and contrast laboratory findings, enzyme deficiency, clinical manifestations and management of Essential Fructosuria and Hereditary Fructose Intolerance. |
| SOM. MK.I.BPM2.3.DM.2.BCHM.1169 | Describe the polyol pathway, its function and its importance in diabetic patients. |
| SOM. MK.I.BPM2.3.DM.2.BCHM.1170 | Describe the metabolism and fate of dietary galactose. |
| SOM. MK.I.BPM2.3.DM.2.BCHM.1171 | Compare and contrast laboratory findings, enzyme deficiency, clinical manifestations and management of Classical and non-classical galactosemia. |
| SOM. MK.I.BPM2.3.DM.2.BCHM.1172 | Differentiate benign fructosuria, galactosemia, lactose intolerance and hereditary fructose intolerance based on clinical features, laboratory findings and dietary management. |

# Recommended Reading

- Lippincott's Biochemistry 9th Edition
  🔗 https://sgu.idm.oclc.org/login?url=https://premiumbasicsciences.lwwhealthlibrary.com/book.aspx?bookid=3402
  - Chapter 12: Monosaccharide and Disaccharide Metabolism (Concept map Fig: 12.8)
- Monosaccharide and Disaccharide Metabolism | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library
- Lippincott's Q: 12.1 - 12.5

# Reducing sugar in urine

- An 8-month-old child has presence of reducing sugar in urine. Urine sample is positive for dipstick test with Clinitest.
  - Reducing sugars convert cupric ions to cuprous ions
- What disorders do you think of in this clinical situation?
- What other clinical signs and symptoms are present?


Galactose



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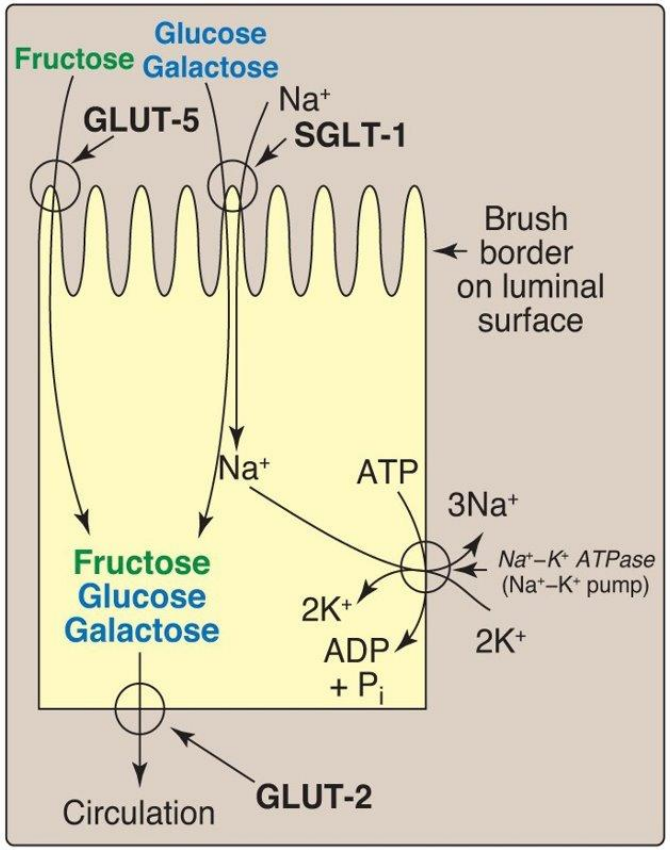
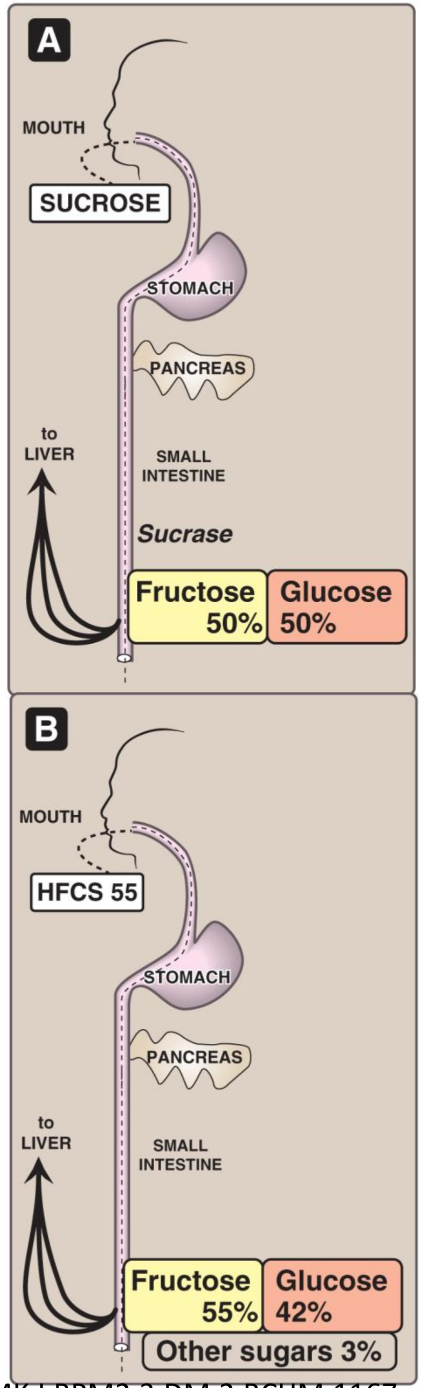
# Fructose metabolism

- Dietary sources of **fructose**
  - Sucrose – Digested by sucrase
  - Fruits and honey
  - High fructose corn syrup (55% fructose/ 45% glucose)
  - Sorbitol (forms fructose by sorbitol dehydrogenase)
- Absorbed by **GLUT-5** (facilitated diffusion)
- Metabolized in Liver
- Fructokinase
- Aldolase B in liver (Same aldolase B of glycolysis)
  - Higher affinity for Fructose 1,6 bisphosphate (glycolysis) than for Fructose 1-phosphate (fructose metabolism)



# High fructose corn syrup (HFCS)

- Sweetener in beverages, ….
- Higher incidence of obesity
- Limit dietary HFCS content
- Processed foods and beverages



# Fructose metabolism

**Aldolase B:** Higher affinity for Fructose 1,6-bisP (glycolysis) than for Fructose 1-P (fructose metabolism)

Impairs **fructose metabolism**; glycolysis and gluconeogenesis

**Fructose**

**Fructokinase**

**Fructose 1-phosphate**

**Aldolase B**

**Glyceraldehyde**

**Dihydroxyacetone phosphate**

**Glyceraldehyde 3-phosphate**

**Intermediates of glycolysis**

# Case report

- An 18-year-old girl avoids eating fruits and table sugar
- **Severe weakness** and **hypoglycemia** (Remember: hypoglycemia a few hours after sucrose or fructose, **trigger**!!!)
- **Drowsy** and **apathetic** (symptoms of hypoglycemia)
- Mom eliminated these foods
- Elimination of foods (trigger), symptoms less frequent
- Name foods eliminated by Candice's mom
- Is this Essential fructosuria or Hereditary fructose intolerance?

# Benign fructosuria (Essential fructosuria)

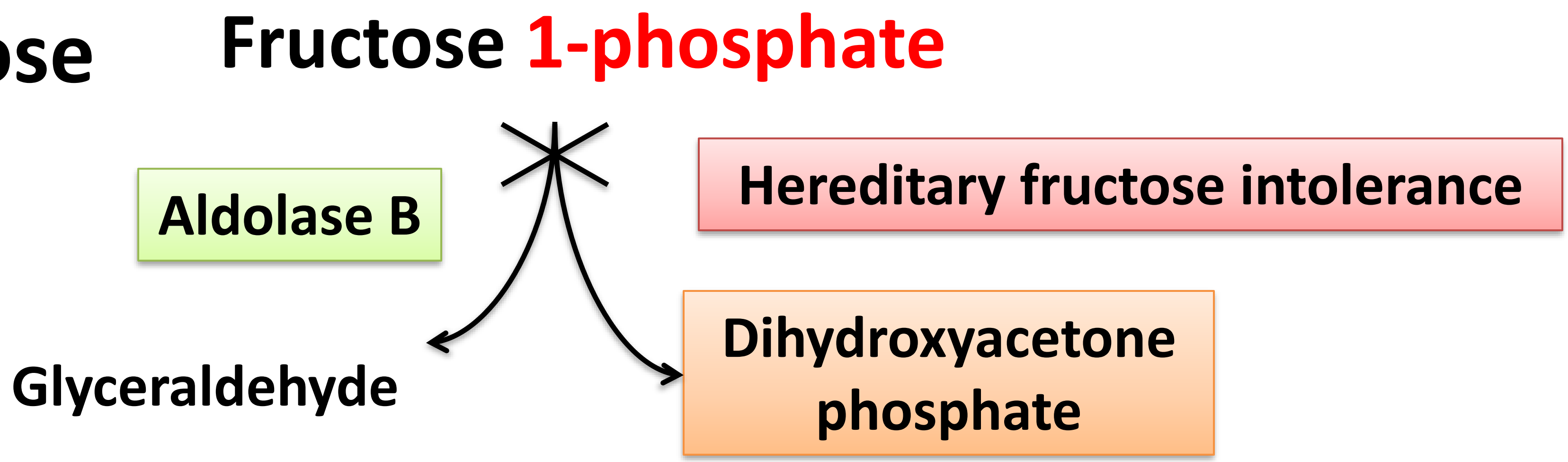
Fructose

**Fructokinase** ↓ **Essential fructosuria**

Fructose 1-phosphate

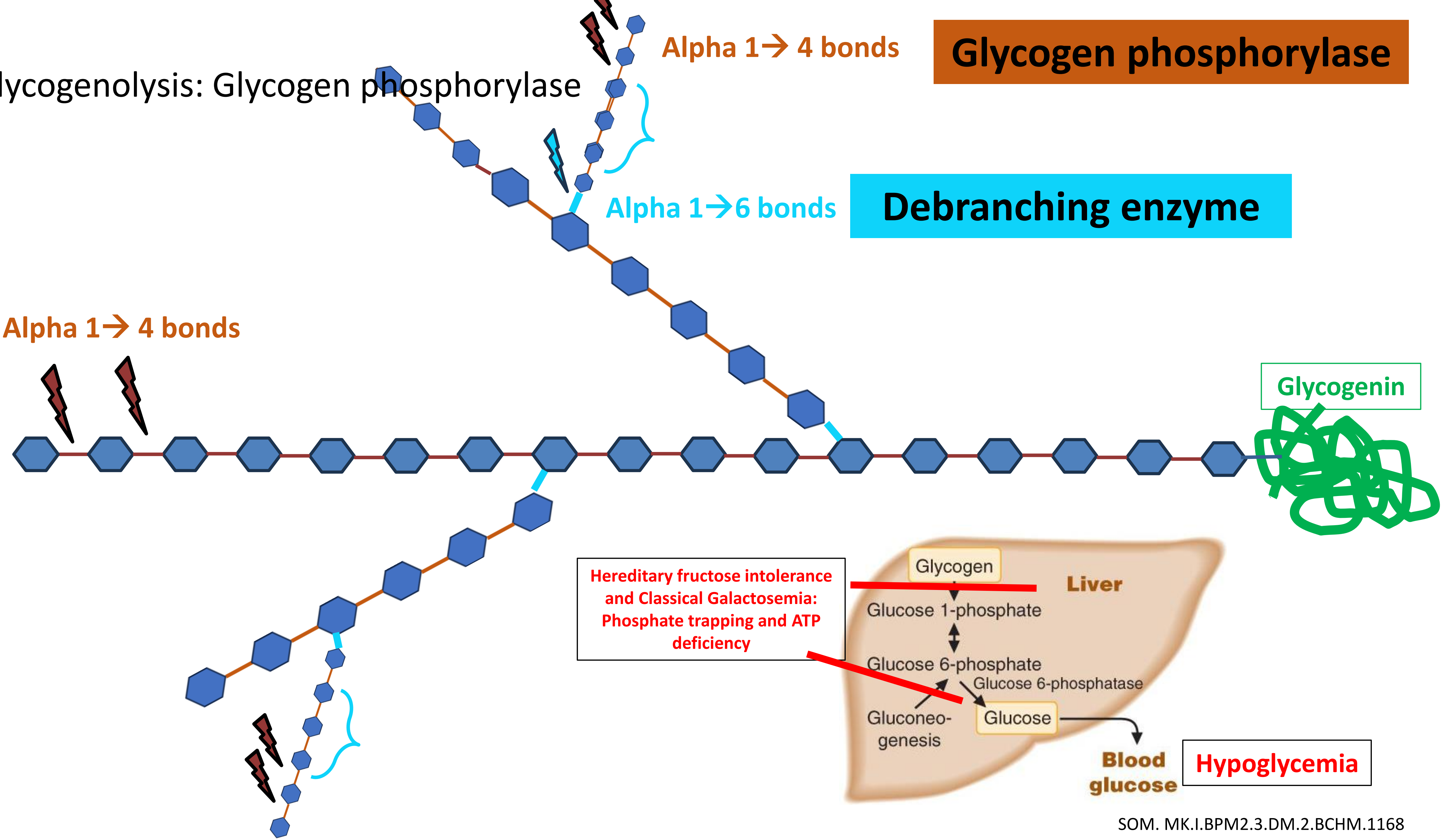
- Liver **fructokinase** deficiency
- Fructose NOT metabolized and excreted in urine
- No toxic metabolites accumulate – **Asymptomatic**
- Urinalysis shows reducing sugar, that is NOT glucose or galactose
- Detected by abnormal urinalysis report

# Hereditary fructose intolerance




- **Liver Aldolase B** deficiency
- Dietary sucrose/ fructose→ **Fructose 1-phosphate** accumulates
- Trapping of inorganic phosphate (Pi)
  - results in ATP deficiency→ inhibits gluconeogenesis
  - Inhibits glycogenolysis (Glycogen phosphorylase requires inorganic phosphate)
- Leads to hypoglycemia (drowsy and apathetic)
- Remember, hypoglycemia **following sucrose or fructose** (**trigger**)
- Symptoms after weaning and **fruits** added (6-8 months of life)

# Glycogenolysis: Glycogen phosphorylase



# Hereditary fructose intolerance

- Continued dietary fructose/ sucrose/ sorbitol → **hepatocellular failure** and jaundice (Elevated AST and ALT)
- Urinalysis shows reducing sugar (fructose) that is NOT glucose
- Children with fructose intolerance do NOT have cataracts (Remember!!)
- Avoid dietary fructose→ good dental health and NO dental caries!!
- Have aversion to sweets!!
- **Diagnosis: Hypoglycemia following sucrose/ fructose in diet, reducing sugar in urine and enzyme assays**

# Synthesis of fructose from glucose (polyol pathway)

- Fructose can be synthesized from sorbitol (Children with fructose intolerance avoid sorbitol)
- **Sorbitol** (sugar alcohol) from glucose by aldose reductase
- **Aldose reductase** rich in lens, retina, peripheral nerves and seminal vesicles
- Polyol pathway forms fructose in seminal vesicle – rich in sorbitol dehydrogenase (fructose: Main fuel for spermatozoa)

**Glucose**

$\downarrow$ **Aldose reductase**

**Sorbitol**

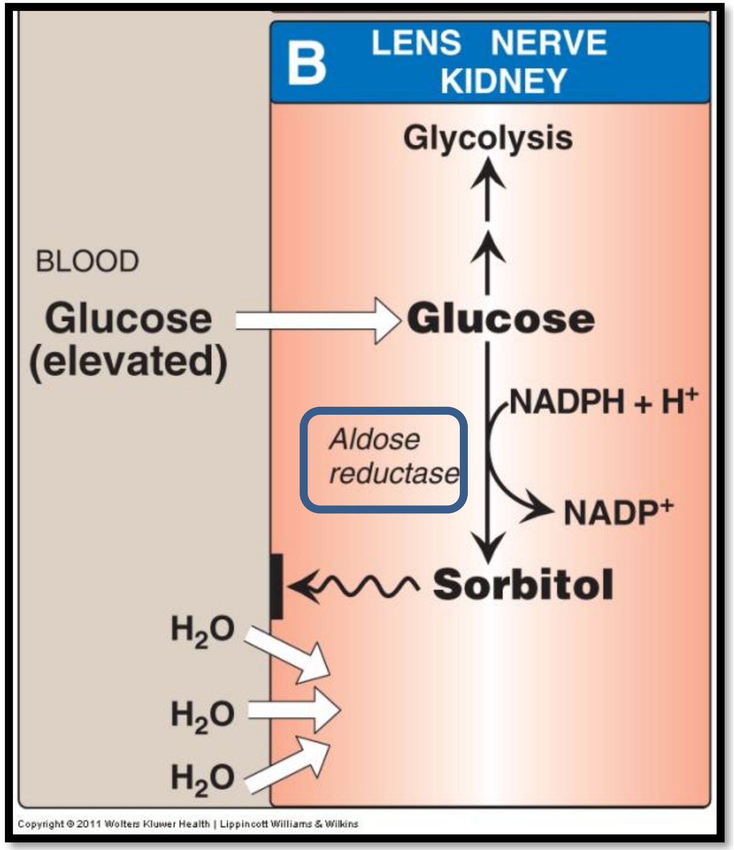
$\downarrow$ **Sorbitol dehydrogenase**

**Fructose**

# Significance of polyol pathway

**In uncontrolled diabetes mellitus,**

- Plasma glucose, enters lens and forms sorbitol by aldose reductase
- Sorbitol (osmotically active) increases water content of lens, results in cataract (decreased lens transparency)
- Sorbitol is partly implicated in pathogenesis of microvascular complications of **diabetes mellitus** (neuropathy, nephropathy, retinopathy) – Insulin independent glucose uptake

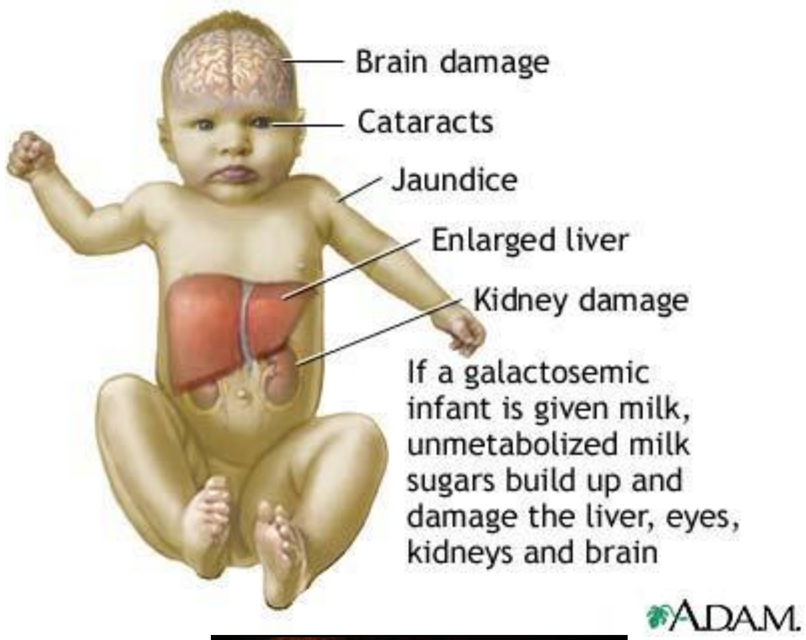


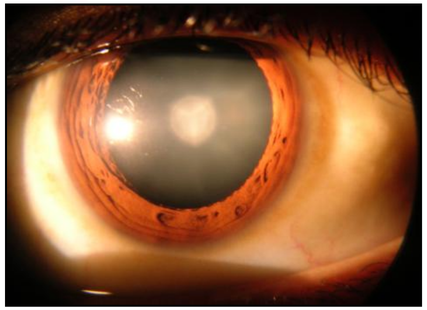
# Significance of polyol pathway

- In **galactosemia**,
  - Elevated blood galactose →entry of galactose into lens
  - Galactose → **Galactitol** by **aldose reductase**
  - Galactitol (osmotically active) increases water content of lens → cataract (Lens opacity)
  - Infants with **untreated galactosemia** have cataracts

# Case report on galactosemia

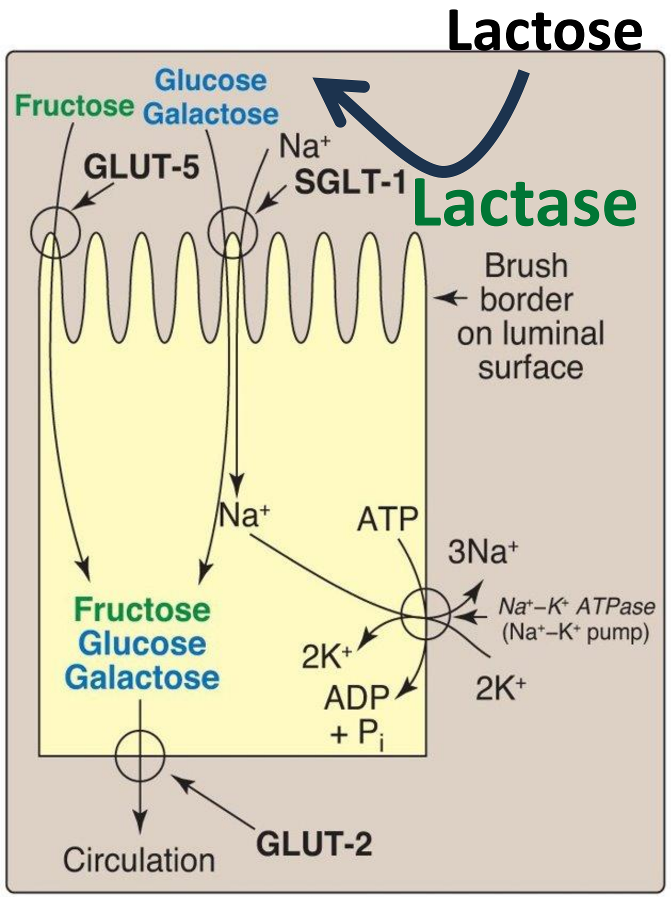
- A 3-week-old infant has **worsening jaundice**
- Vomiting after feeds and failure to thrive
- **Higher risk of sepsis (E.coli)**
- Liver enlarged (**hepatomegaly**)
- Early **cataract** formation
- **Developmental milestone delay**
- **Classical triad: liver damage, developmental delay and cataracts**
- Urine positive for **reducing sugar** (not glucose)
- Inborn error of galactose metabolism???



# Galactose metabolism

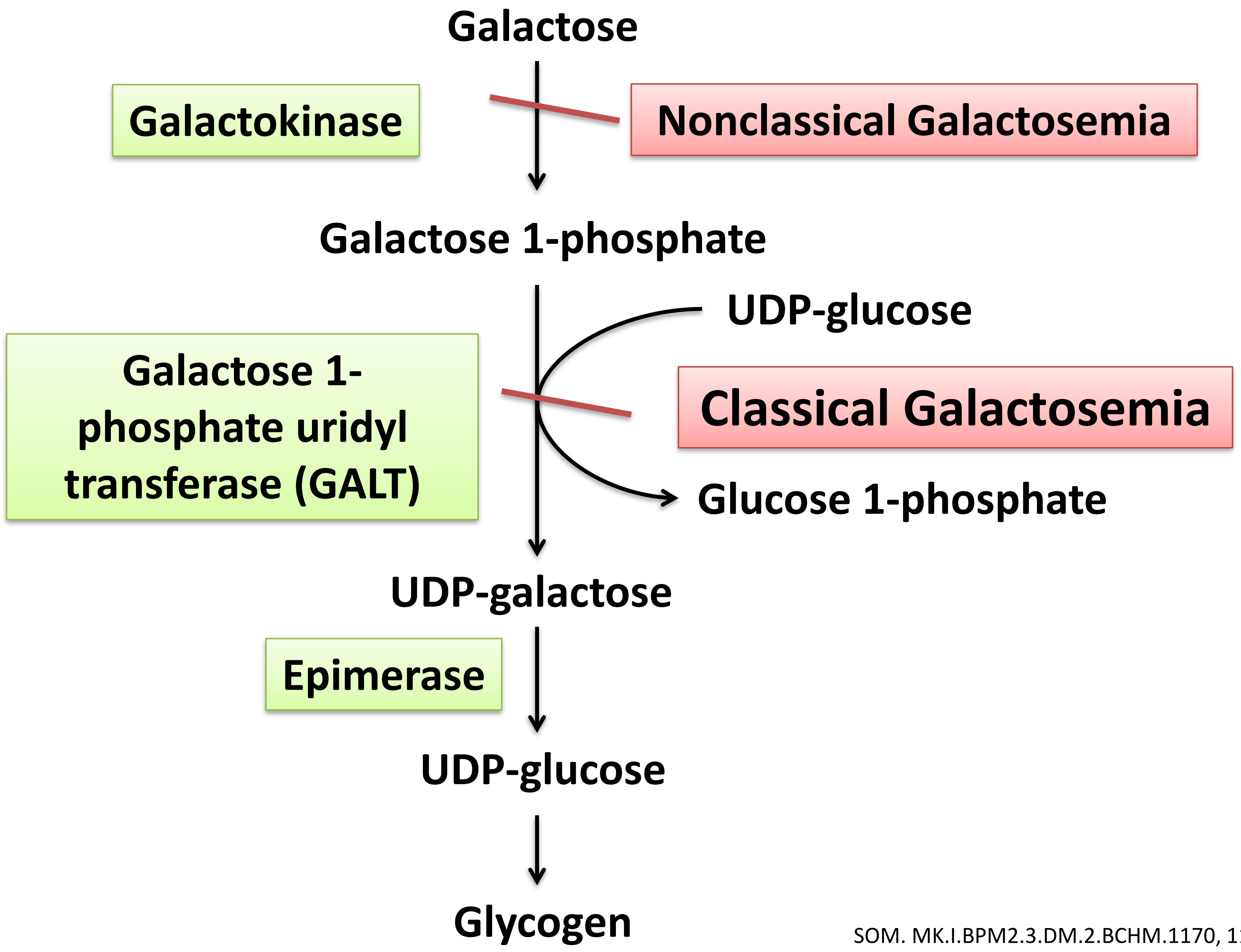
- Lactose (milk sugar) (β1→4 linkage)
  - Intestinal lactase to glucose and galactose
- **SGLT-1** (Sodium dependent glucose transporter-1)
- Metabolized in Liver
- Galactokinase, **galactose 1-phosphate uridyl transferase** (GALT) and epimerase
- Galactose eventually converted to glucose (glycogen)





- **Lactose intolerance**: Deficiency of intestinal lactase (disaccharidase) → **Diarrhea, bloating, cramps**; No reducing sugar in urine; no liver involvement/ cataracts/developmental delay

# Galactose metabolism

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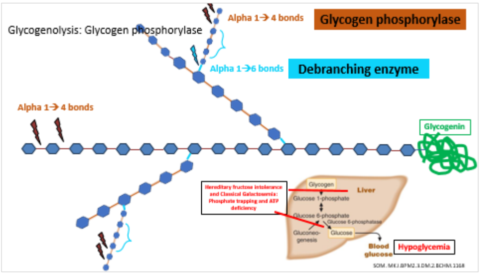
# Classical galactosemia




- Galactose 1-phosphate uridyl transferase deficiency (GALT)
- **Autosomal recessive**
- Galactose and **galactose 1-phosphate** accumulate
- Presentation: **2nd -3rd week of life**
- **Higher risk of sepsis in neonates**
- **Galactose 1-phosphate** → trapping of phosphate → decreased gluconeogenesis and glycogenolysis →**Hypoglycemia**
- Remember, hypoglycemia after dietary lactose (milk) or galactose (**trigger**!!!)
- **Galactose 1-phosphate** and **galactitol** accumulation→ **Jaundice and hepatomegaly** (Elevated AST and ALT)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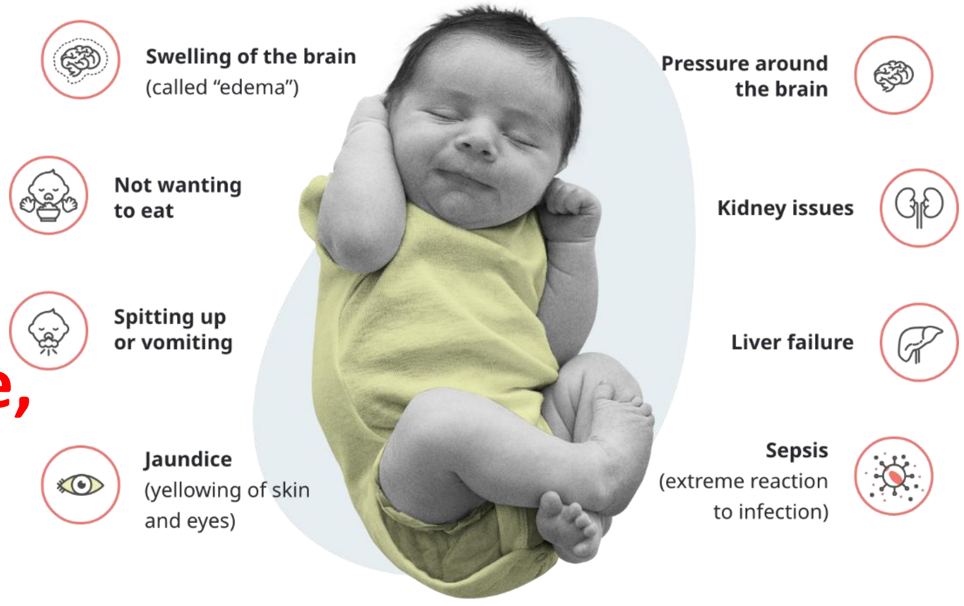
# Classical galactosemia

- Lens: Galactose → **galactitol** (**aldose reductase**)
- Galactitol increases **lens** water content→ **cataracts**
- **Galactose 1-phosphate and galactitol** accumulate in **brain** cause **neurological damage,** learning disability and milestone delay
- Newborns screening: Heel prick test
  - Measure Galactose or galactose 1-phosphate or GALT activity
- Urinalysis: **Reducing sugar** (galactose) that is NOT glucose

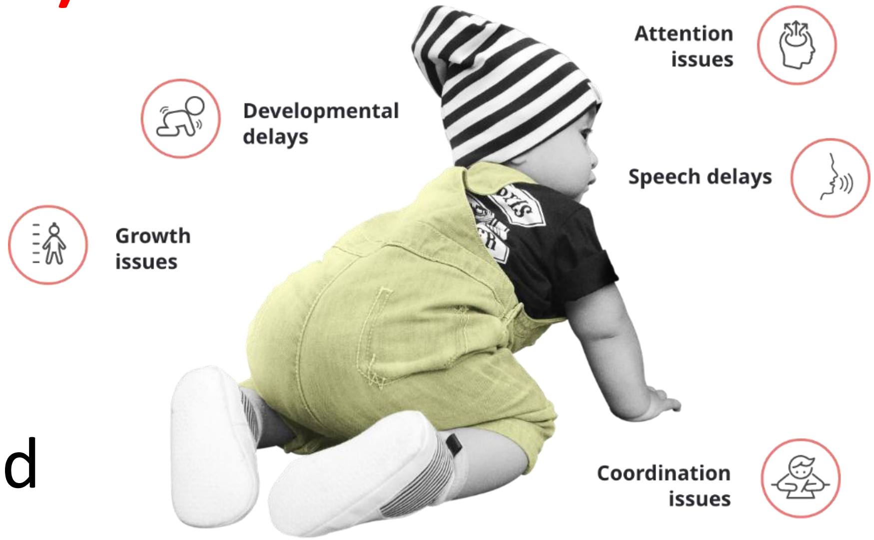



Newborn

https://www.youtube.com/watch?v=Q7oEz6pmhPA

# Management of galactosemia

- Dietary exclusion of galactose/ lactose
  - Improves liver function and cataracts
  - Effect on learning disability limited and varied
- **Breast feeding stopped**
- Substitute **soy-milk or lactose-free milk (not Lactaid)**
- Lifelong dietary restrictions
- Monitor Galactose 1-phosphate levels
- Females with treated galactosemia, have delayed puberty and premature ovarian failure
- Many treated children also have developmental and speech problems

# Non-classical galactosemia

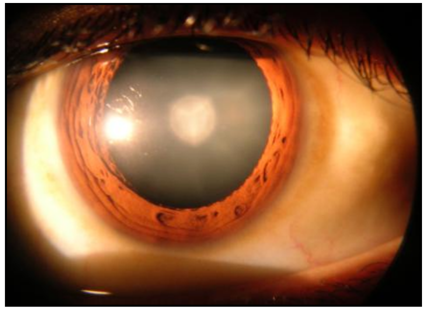
Galactose

Galactokinase | Nonclassical Galactosemia

↓

Galactose 1-phosphate

- Galactokinase deficiency
- **Lens Galactitol** formation → **Cataracts**
- NO galactose 1-phosphate accumulation
- Less severe than classical galactosemia
- Urine positive for **reducing sugar** (galactose)
- Monitor Galactitol levels
- Dietary lactose/ galactose restriction



# Lactose intolerance

| | Enzyme deficient and accumulated substrate | Predominant clinical features | Biochemical basis for clinical features | Biochemical basis for management |
|---|---|---|---|---|
| **Lactose intolerance** | Intestinal Lactase (disaccharidase) | Diarrhea, bloating (**No** reducing sugar in urine; **No** cataracts, liver damage, or developmental delay) | Intestinal undigested lactose is degraded by bacterial action into gases | Lactose avoidance in diet |

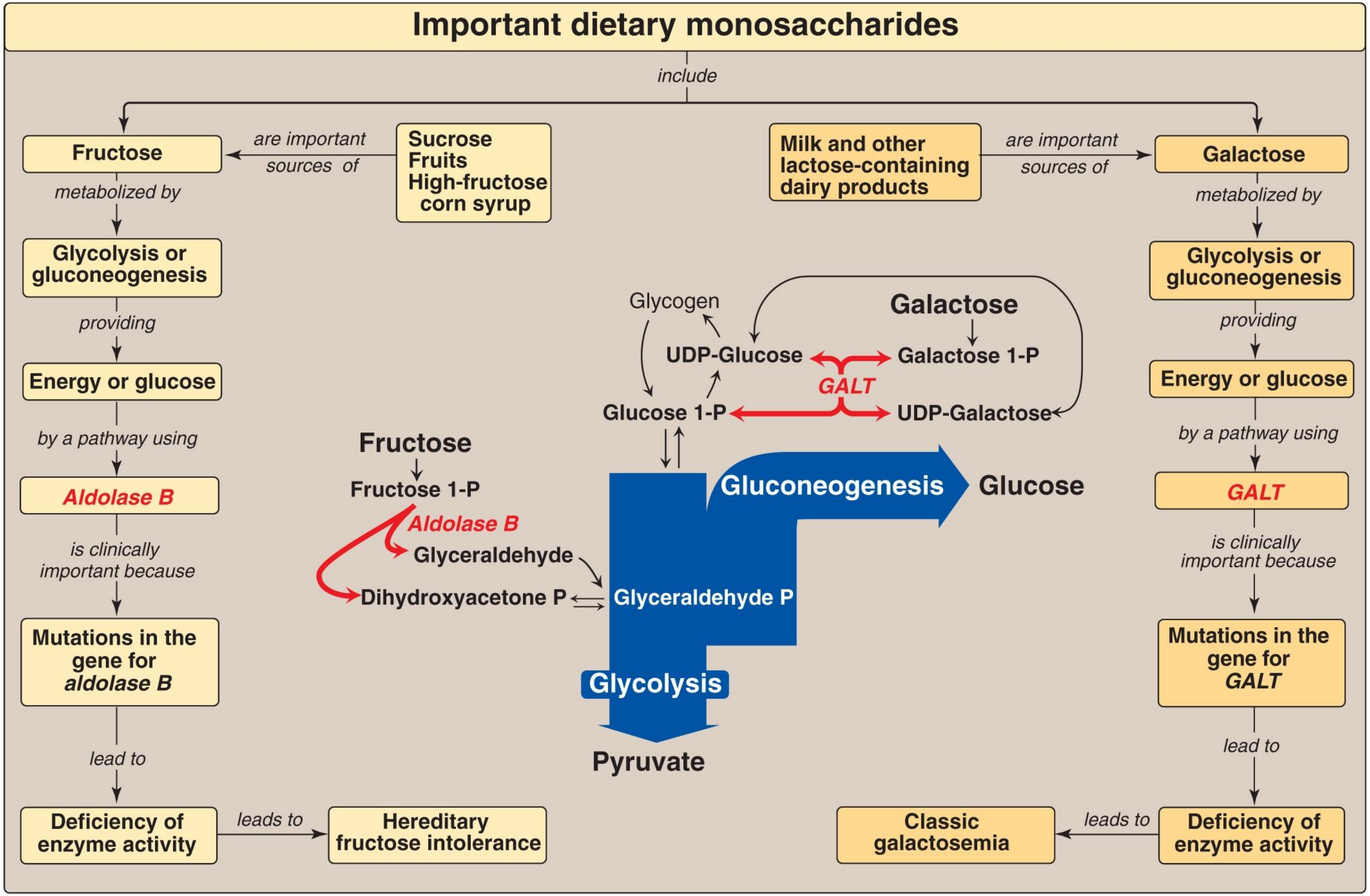
# Reducing sugar in urine: Differential diagnosis

- An 8-month-old child has a positive dipstick test with Clinitest.
- What disorders do you think of in this clinical situation?
  - Inherited disorders of Fructose or Galactose metabolism
- What other clinical signs and symptoms are present?
  - Cataracts, liver disease, developmental delay **(Classical galactosemia)**
  - Cataracts (**Non-classical galactosemia)**
  - Hypoglycemia and liver disease **(Classical galactosemia or hereditary fructose intolerance**
  - Hypoglycemia after sucrose **(hereditary fructose intolerance)** or lactose ingestion **(GALT deficiency)**
  - Benign (No significant clinical features)- **(Benign fructosuria)**

# Compare galactosemia, lactose intolerance, essential fructosuria, hereditary fructose intolerance

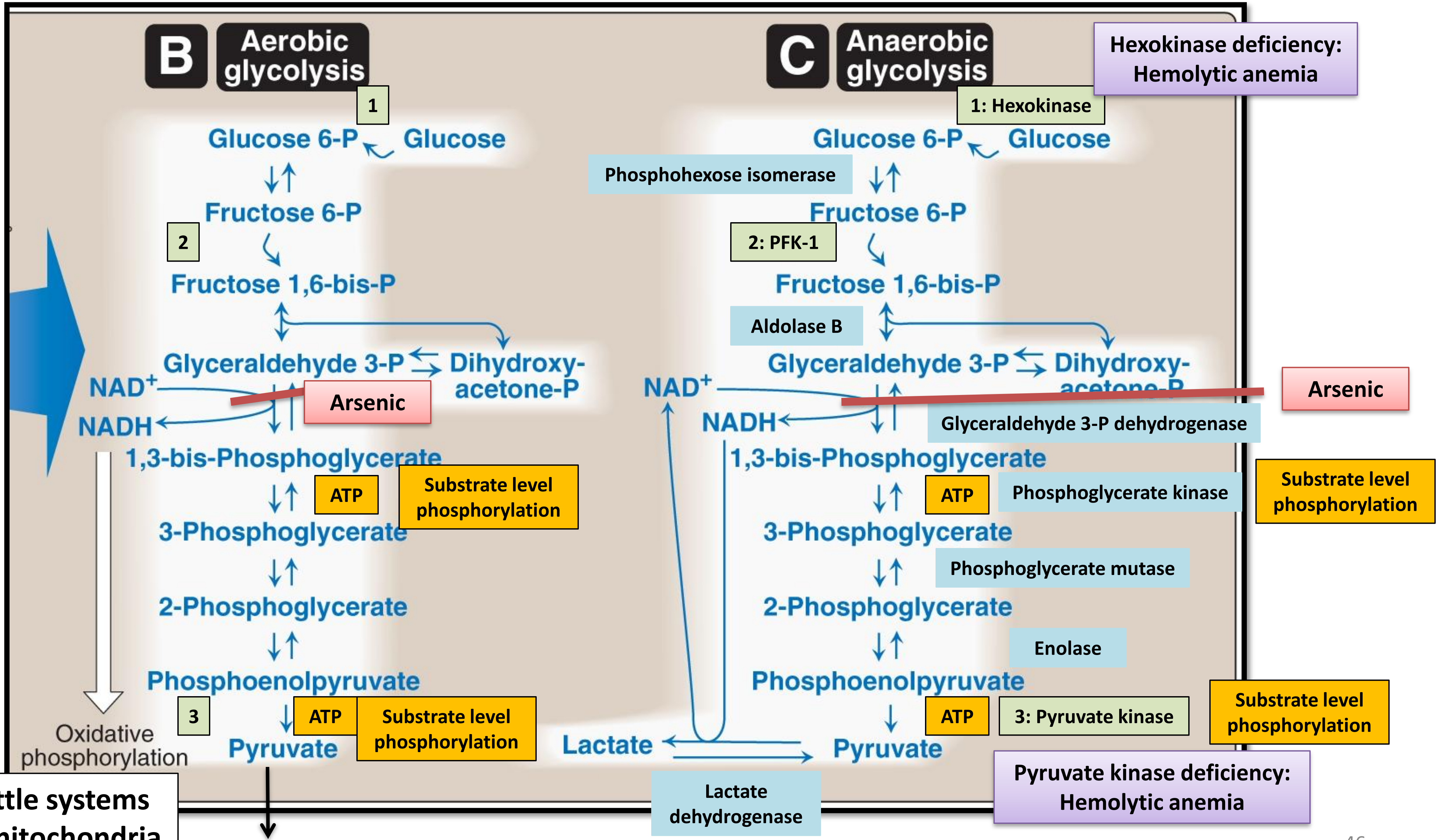
| | Enzyme deficient and accumulated substrate | Predominant clinical features | Biochemical basis for clinical features | Biochemical basis for management |
|---|---|---|---|---|
| **Classical Galactosemia** | | | | |
| **Non-classical Galactosemia** | | | | |
| **Lactose intolerance** | | | | |
| **Essential fructosuria** | | | | |
| **Hereditary fructose intolerance** | | | | |

**Answers in Practice Questions**





Lippincott's fig 12.8




Via Shuttle systems enters mitochondria

**Acetyl CoA**